INDIAN SPACE OLYMPIAD

2026 MOCK QUESTION PAPERS

INTERMEDIATE LEVEL

Classes VII-VIII Practice Set

5 Mock Papers | 50 Questions Each | 250 Questions

Strictly aligned to the ISO Intermediate Level syllabus for Classes VII-VIII,with 10 Olympiad Challenge questions in every paper.

Student name	
School	
Prepared for	Indian Space Olympiad practice

Difficulty design: 40 core questions + 10 Olympiad Challenge questions in every paper.

Indian Space School

How to Use These Mock Papers

These mock papers are written for the ISO Intermediate Level syllabus shared by Classes VII and VIII. Questions 1-40 in each paper test core concepts and applications across the eleven syllabus units. Questions 41-50 form the Olympiad Challenge and are intentionally more demanding, requiring calculations, interpretation, experimental reasoning or multi-step application.

Each paper has 50 multiple-choice questions. Questions 1-40 form the Core Section. Questions 41-50 form the Olympiad Challenge. The challenge section is suitable for strong Class VII students and regular Class VIII Olympiad preparation.

Suggested time: 90 minutes per paper. Suggested marking: 1 mark for every correct answer and no negative marking during practice. Students should attempt the paper independently before checking the answer key and challenge explanations.

Mission-related facts were checked against official ISRO and NASA material available through 23 July 2026. The papers remain strictly within the supplied Intermediate Level syllabus.

Syllabus Coverage

Fundamentals of Astronomy

The Solar System

Stars, Galaxies and the Universe

Earth and Moon

Satellites and Space Technology

Rockets and Launch Vehicles

ISRO and India's Space Missions

Space Exploration

Basic Physics for Space Science

Scientific Reasoning and Logical Aptitude

General Space Awareness

Instructions for Students

Choose the best answer from A, B, C and D.

Use rough work for calculations.

Study every figure carefully before answering figure-based questions.

Do not spend too long on one question; return to it later.

Attempt the Olympiad Challenge even when you are unsure.

MOCK PAPER 1

ISO 2026 Mock Paper - Intermediate Level (Classes VII-VIII)

Time: 90 minutes   Maximum Marks: 50

Name	Date	Score	Time taken
		/50	

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question. Suggested time: 90 minutes.

1. Which early Indian astronomer explained that the apparent daily motion of the sky is related to Earth's rotation?

A. Neil Armstrong   B. Isaac Newton

C. Edwin Hubble   D. Aryabhata

2. A model in which Earth is placed at the centre is called the

A. heliocentric model   B. geocentric model

C. nebular model   D. expanding model

3. The imaginary sphere on which stars appear to lie is called the

A. celestial sphere   B. atmosphere

C. asteroid belt   D. magnetosphere

4. The point in the sky directly above an observer is the

A. horizon   B. zenith

C. ecliptic   D. north pole

5. In the altitude-azimuth system, altitude tells us

A. the age of a galaxy

B. the distance of a planet from the Sun

C. how high an object is above the horizon

D. the colour of a star

6. A light-year is a unit of

A. distance   B. mass

C. brightness   D. time

7. A refracting telescope mainly uses a large

A. magnet   B. mirror

C. radio dish   D. lens

8. A telescope with a larger aperture usually collects

A. no radio waves   B. only red light

C. more light   D. less light

9. Which radiation has the longest wavelength in the electromagnetic spectrum?

A. X-rays   B. Visible light

C. Radio waves   D. Gamma rays

10. Which detail is most important to record during a sky observation?

A. Only the observer's name

B. The colour of the notebook

C. The price of the telescope

D. Time, place and direction

11. According to the Solar Nebula Theory, the Solar System formed from a rotating cloud of

A. only rocks B. dark matter

C. gas and dust D. liquid water

12. Which group contains only terrestrial planets?

A. Mercury, Jupiter, Uranus, Neptune

B. Mercury, Venus, Earth, Mars

C. Jupiter, Saturn, Uranus, Neptune

D. Earth, Jupiter, Saturn, Mars

13. Which planet is a giant planet?

A. Venus B. Jupiter

C. Mercury D. Mars

14. A comet's tail generally points

A. towards Earth

B. towards the Sun

C. away from the Sun

D. in the direction of its motion only

15. The Kuiper Belt lies mainly

A. inside the Sun B. between Mars and Jupiter

C. beyond Neptune D. between Earth and Mars

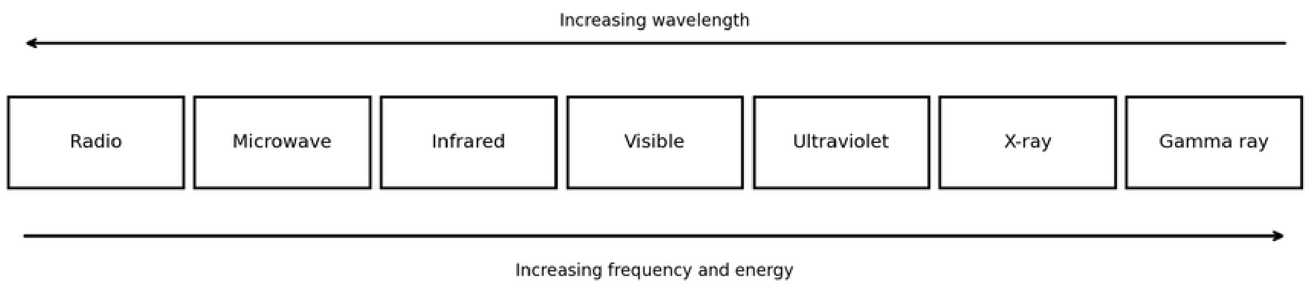

Figure 1. Simplified electromagnetic spectrum.

16. Study Figure 1. Visible light lies between

A. microwave and radio B. radio and microwave

C. X-ray and gamma ray D. infrared and ultraviolet

17. Study Figure 1. Which radiation has a higher frequency than ultraviolet?

A. Infrared B. X-ray

C. Radio D. Microwave

18. Study Figure 1. A large dish antenna is most directly associated with detecting

A. visible light only B. gamma rays only

C. radio waves D. sound waves

19. A main-sequence star produces energy mainly by

A. nuclear fusion in its core B. falling through space

C. burning coal D. reflecting moonlight

20. A Sun-like star becomes a red giant after it begins to run low on

A. hydrogen in its core B. oxygen in its atmosphere

C. water vapour D. iron on its surface

21. The compact remnant of a Sun-like star is usually a

A. black hole B. comet

C. neutron star D. white dwarf

22. The Milky Way is best described as a

A. planet B. single nebula

C. spiral galaxy D. star cluster inside the Moon

23. A black hole is an object whose gravity is so strong that

A. even light cannot escape from within its event horizon

B. it has no mass

C. it produces sunlight

D. it stops the Universe from expanding

24. The thin, solid outer layer of Earth is the

A. crust B. outer core

C. inner core D. mantle only

25. Earth's magnetosphere helps protect us from much of the

A. solar wind B. ocean tide

C. rainfall D. moonlight

26. The dark plains visible on the Moon are called

A. auroras B. maria

C. coronae D. continents

27. A solar eclipse occurs when

A. the Moon is between the Sun and Earth

B. the Sun is between Earth and Moon

C. Mars blocks the Sun

D. Earth is between the Sun and Moon

28. Spring tides are generally strongest near

A. a crescent Moon only B. new Moon and full Moon

C. first and third quarter Moon only D. no Moon phase

29. A geostationary satellite appears fixed above one point because it

A. is attached to Earth by a cable

B. orbits Earth once in about the same time Earth rotates

C. orbits over both poles

D. does not move

30. A polar-orbiting satellite is especially useful for

A. making stars brighter

B. observing nearly the whole Earth over time

C. landing on the Moon

D. remaining above one city

31. In a communication satellite, a transponder mainly

A. creates gravity B. cools the Sun

C. receives and retransmits signals D. makes rocket fuel

32. Attitude control keeps a spacecraft

A. moving faster than light

B. pointed in the required direction

C. free from all gravity

D. inside Earth's atmosphere

33. A rocket rises because gases are pushed downward and the rocket is pushed upward. This is an example of

A. Archimedes' principle B. Newton's first law only

C. reflection of light D. Newton's third law

34. Why are launch vehicles built in stages?

A. To make them heavier throughout flight

B. To stop them entering orbit

C. To discard empty mass and improve efficiency

D. To avoid using engines

35. PSLV is widely known for launching many satellites into

A. underground laboratories B. the surface of the Sun

C. ocean currents D. polar and Sun-synchronous orbits

36. Aditya-L1 is designed mainly to study the

A. Jupiter B. Mars

C. Moon D. Sun

37. XPoSat studies

A. X-ray polarisation from cosmic sources

B. ocean salinity only

C. plant growth on the Moon

D. sound waves from stars

38. The International Space Station is mainly used as a

A. weather balloon B. radio telescope on Earth

C. launch pad on the Moon D. microgravity research laboratory

39. A force of 20 N moves a box through 3 m in the direction of the force. The work done is

A. 17 J B. 60 J

C. 6 J D. 23 J

40. Momentum depends on an object's

A. mass and velocity B. shape only

C. volume only D. colour and temperature

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately difficult. Read carefully, calculate where needed and use elimination.

41. A radio signal takes 1.3 seconds to travel from Earth to the Moon. Approximately how long will a reply take to travel from Earth to the Moon and back?

A. 13 s B. 1.3 s

C. 2.6 s D. 0.65 s

42. A star is observed 30 degrees above the eastern horizon. Which pair correctly describes its position?

A. Altitude 30 degrees; direction east

B. Declination 30 km; direction west

C. Altitude 60 degrees; direction north

D. Altitude east; direction 30 degrees

43. Planet A is closer to the Sun than Planet B. If both follow ordinary planetary orbits, which statement is most likely correct?

A. A usually takes longer to orbit

B. Both must take exactly one year

C. Distance has no relation to orbital period

D. B usually takes longer to orbit

44. A force of 120 N acts on an area of 0.03 m². What pressure is produced?

A. 4 Pa B. 40 Pa

C. 400 Pa D. 4000 Pa

45. A satellite sends the following data: 08:00 - 20 units, 09:00 - 35 units, 10:00 - 35 units, 11:00 - 50 units. Between which two observations was there no change?

A. 09:00-10:00 B. 08:00-09:00

C. 08:00-11:00 D. 10:00-11:00

46. Solar eclipses do not occur at every new Moon mainly because

A. the Moon produces its own light

B. the Moon's orbit is tilted relative to Earth's orbit

C. the Sun changes direction each month

D. Earth stops rotating

47. An astronaut has a mass of 50 kg on Earth. On the Moon, the astronaut's

A. mass is less but weight is the same

B. mass and weight are both greater

C. mass is the same but weight is less

D. mass and weight are both zero

48. Two telescopes have equal-quality optics. Telescope X has twice the aperture diameter of Telescope Y. Telescope X will mainly have

A. greater light-collecting ability and finer detail

B. exactly the same performance in every way

C. less light-collecting ability

D. no ability to observe faint objects

49. A student tests whether the colour of a surface affects heating in sunlight. Which variables should be kept the same?

A. The student's opinion

B. Material, size, exposure time and starting temperature

C. Final temperature only

D. Surface colour only

50. Complete the pattern: star, planet, planet, star, planet, planet, star, ___, ___.

A. Moon, comet B. planet, star

C. star, star D. planet, planet

MOCK PAPER 2

ISO 2026 Mock Paper - Intermediate Level (Classes VII-VIII)

Time: 90 minutes Maximum Marks: 50

Name	Date	Score	Time taken
		/50	

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question. Suggested time: 90 minutes.

1. Galileo improved astronomy by using a telescope to observe

A. the inside of Earth

B. the centre of a black hole

C. sound waves from Mars

D. mountains on the Moon and moons of Jupiter

2. In basic celestial coordinates, declination is most similar to

A. longitude on Earth B. latitude on Earth

C. speed of a rocket D. height of a mountain

3. Azimuth is measured along the

A. axis of a telescope B. horizon

C. Moon's diameter D. line from Earth to the Sun only

4. Parallax can help astronomers estimate the distance to

A. sound sources B. all galaxies equally well

C. only the Sun's surface D. nearby stars

5. One astronomical unit is approximately the average distance between

A. Earth and Sun B. Mars and its moons

C. Earth and Moon D. Sun and Jupiter

6. A reflecting telescope uses a curved

A. glass prism only B. radio transmitter

C. thermometer D. mirror

7. The main purpose of a telescope mount that tracks the sky is to

A. move the stars

B. increase the speed of light

C. keep an object in view as Earth rotates

D. create a larger Moon

8. Infrared observations are useful for studying some cool objects and dusty regions because infrared radiation can

A. be seen directly by the eye

B. pass through some dust better than visible light

C. travel slower than sound

D. exist only on Earth

9. Which measurement is most suitable for the diameter of a planet?

A. Kelvin per hour B. Seconds

C. Kilometres D. Newtons

10. If a star appears brighter through a telescope than to the unaided eye, the telescope has mainly collected

A. more gravity B. more time

C. more sound D. more light

11. As the Solar Nebula contracted, it rotated faster mainly because of conservation of

A. angular momentum B. pressure only

C. electric charge only D. colour

12. Mercury has difficulty keeping a thick atmosphere mainly because it is

A. made only of gas B. small and hot

C. very large and cold D. beyond Neptune

13. Venus is extremely hot mainly because of its strong

A. distance beyond Jupiter   B. ocean currents

C. greenhouse effect   D. magnetic storms only

14. Mars has an atmosphere that is mostly

A. carbon dioxide   B. oxygen

C. helium   D. hydrogen

15. Uranus and Neptune are often called

A. dwarf planets   B. ice giants

C. terrestrial planets   D. asteroids

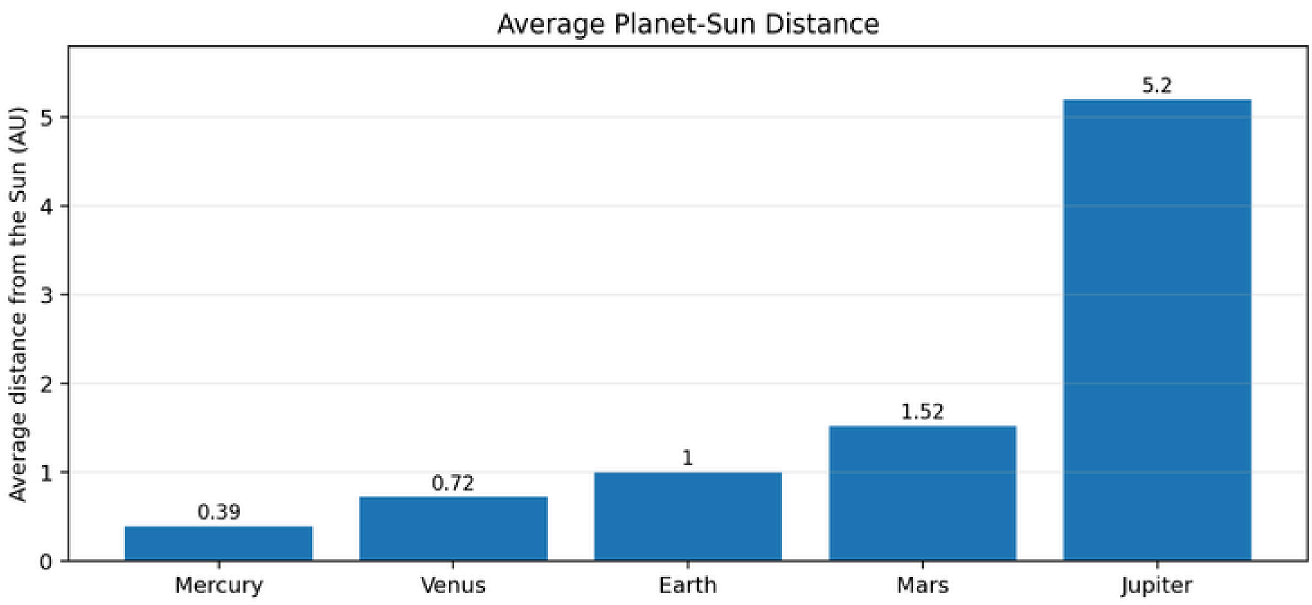

Figure 2. Average distance of selected planets from the Sun.

16. Study Figure 2. Which planet is shown at 1 AU?

A. Earth   B. Venus

C. Mars   D. Jupiter

17. Study Figure 2. Which planet is farthest from the Sun among those shown?

A. Jupiter   B. Earth

C. Mercury   D. Mars

18. Study Figure 2. Mars is approximately how many AU farther from the Sun than Earth?

A. 0.12 AU   B. 4.20 AU

C. 1.52 AU   D. 0.52 AU

19. Most asteroids in the main asteroid belt orbit between

A. Mars and Jupiter B. Earth and Moon

C. Saturn and Uranus D. Mercury and Venus

20. A Near-Earth Object is an asteroid or comet whose orbit

A. is always beyond Neptune B. never approaches any planet

C. comes close to Earth's orbit D. lies inside the Sun

21. Blue stars are generally

A. the same temperature as all red stars

B. not real stars

C. cooler than red stars

D. hotter than red stars

22. A group of stars formed from the same cloud and held together by gravity is a

A. planetary atmosphere B. star cluster

C. comet tail D. solar flare

23. A supernova can occur at the end of the life of a

A. communication satellite B. small asteroid

C. Moon crater D. massive star

24. A neutron star contains matter packed to extremely high

A. density B. volume

C. temperature only at its surface D. transparency

25. The observation that many distant galaxies are moving away supports the idea of an

A. Earth-centred galaxy B. unchanging Universe of one star

C. ocean-centred Solar System D. expanding Universe

26. The ozone layer is mainly found in the

A. troposphere only at ground level B. stratosphere

C. outer core D. lithosphere

27. Earth's liquid metallic layer is the

A. upper atmosphere B. crust

C. inner core D. outer core

28. Climate describes

A. a satellite orbit   B. the Moon's temperature

C. long-term patterns of weather   D. weather at one moment only

29. The Moon keeps nearly the same face towards Earth because its rotation period is

A. zero   B. about equal to its orbital period

C. much shorter than one hour   D. one day

30. Tides are caused mainly by the gravity of the

A. Moon and Sun   B. Milky Way centre only

C. Mars only   D. asteroid belt

31. GPS satellites mainly operate in

A. geostationary orbit only   B. a lunar orbit

C. inside the atmosphere   D. medium Earth orbit

32. A geostationary communication satellite must orbit above the

A. South Pole   B. equator

C. North Pole   D. Moon

33. Remote sensing means collecting information about an object or area

A. without using any detector   B. without direct physical contact

C. only by touching it   D. only with a ruler

34. A spacecraft power system often uses

A. only heat shields   B. water wheels

C. solar panels and batteries   D. only parachutes

35. Escape velocity is the minimum speed needed to

A. make mass disappear

B. escape a body's gravity without further propulsion in an ideal case

C. enter Earth's core

D. stop all motion

36. Launching eastward from near the equator can use Earth's

A. rotation to provide extra speed

B. magnetism to remove gravity

C. oceans to stop the rocket

D. clouds to create fuel

37. A reusable launch system is designed so that

A. some major parts can fly again

B. all parts remain in orbit forever

C. the payload returns to the launch pad before launch

D. no fuel is needed

38. LVM3 is associated with

A. building ground telescopes

B. measuring classroom pressure

C. studying only ocean tides

D. launching heavy payloads and the Gaganyaan programme

39. Chandrayaan-3 demonstrated a soft landing near the Moon's

A. equator on the far side only B. largest ocean

C. south polar region D. north pole of Earth

40. SpaDeX was designed to demonstrate

A. a telescope on Mars

B. weather prediction using balloons

C. rendezvous and docking of spacecraft

D. mining inside Earth

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately difficult. Read carefully, calculate where needed and use elimination.

41. A rover travels 240 m in 8 minutes. Its average speed is

A. 32 m/min B. 20 m/min

C. 30 m/min D. 1920 m/min

42. Study Figure 2. Jupiter's average distance is about how many times Earth's?

A. About 2 times B. About 10 times

C. About 3 times D. About 5.2 times

43. Two planets have circular orbits. Planet X is twice as far from the Sun as Planet Y. Which is certainly true?

A. X has a larger orbital circumference

B. X has the same orbital path length

C. X has no gravitational attraction to the Sun

D. X must move faster at every point

44. A telescope has an objective focal length of 1000 mm and an eyepiece focal length of 20 mm. Its magnification is

A. 50× B. 100×

C. 20× D. 20000×

45. A metal spoon and a wooden spoon are kept in the same room. The metal spoon feels colder mainly because metal

A. contains no particles

B. has a lower room temperature

C. conducts heat away from the hand faster

D. produces cold energy

46. A 2 kg object moves at 4 m/s. Its momentum is

A. 8 kg m/s B. 0.5 kg m/s

C. 2 kg m/s D. 6 kg m/s

47. A student changes the distance between a lamp and a solar cell and measures current. The dependent variable is the

A. distance from lamp B. type of lamp if unchanged

C. current produced D. size of the table

48. If one GPS satellite fails, a receiver can still work accurately because

A. GPS uses no satellites

B. it can use signals from other satellites

C. the receiver creates new satellites

D. Earth stops rotating

49. Which sequence correctly orders these events in a massive star's life?

A. Main sequence → nebula → white dwarf → comet

B. Supernova → nebula → main sequence → asteroid

C. Nebula → main sequence → red supergiant → supernova

D. Red giant → planet → black hole → protostar

50. A rule says: every observation satellite in a list has a camera; Satellite P is in the list. Which conclusion is valid?

A. P must study only Mars B. P definitely has a camera

C. P is geostationary D. P definitely has astronauts

MOCK PAPER 3

ISO 2026 Mock Paper - Intermediate Level (Classes VII-VIII)

Time: 90 minutes   Maximum Marks: 50

Name	Date	Score	Time taken
		/50	

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question. Suggested time: 90 minutes.

1. Right ascension is commonly measured in

A. newtons   B. hours, minutes and seconds

C. kilograms   D. degrees Celsius only

2. Declination is measured north or south of the

A. Sun's core   B. local horizon only

C. celestial equator   D. surface of the Moon

3. An optical telescope on a high mountain benefits from

A. less access to stars   B. stronger sound waves

C. less atmosphere above it   D. more city light

4. Spectral lines can reveal information about a star's

A. postal address   B. exact age by sight alone

C. chemical composition   D. number of planets with certainty

5. A detector that measures the amount of light received is performing

A. seismology   B. navigation

C. photometry   D. rocketry

6. Which telescope is least affected by clouds if it observes through wavelengths that can pass through them?

A. Some radio telescopes   B. Solar cooker

C. Human eye only   D. Visible-light telescope only

7. A star chart should include direction labels because the sky appears different when viewed towards

A. the inside of a telescope

B. different parts of the horizon

C. the same point at the same time only

D. a closed room

8. Astronomical images often need long exposures mainly to

A. slow down Earth's rotation

B. collect more light from faint objects

C. increase gravity

D. make radio waves visible to the eye

9. The Sun's visible surface is called the

A. photosphere B. chromosphere only

C. magnetosphere D. core

10. The solar wind is a stream of

A. charged particles from the Sun

B. liquid water from the Sun

C. sound waves in empty space

D. rocks from the asteroid belt

11. The atmosphere of Jupiter is made mainly of

A. carbon dioxide only B. oxygen and nitrogen

C. water vapour only D. hydrogen and helium

12. Pluto is classified as a dwarf planet partly because it has not

A. become round B. cleared its orbital neighbourhood

C. formed beyond Neptune D. orbited the Sun

13. A comet becomes most active when it

A. enters Earth's ocean B. stops moving

C. approaches the Sun D. moves far beyond the Kuiper Belt

14. A meteor is the streak of light produced when a small object

A. becomes a star B. lands without heating

C. enters an atmosphere and heats up D. orbits safely beyond Neptune

15. A meteorite is a piece that

A. remains in the asteroid belt

B. reaches the ground

C. becomes a comet tail

D. burns completely before entering the atmosphere

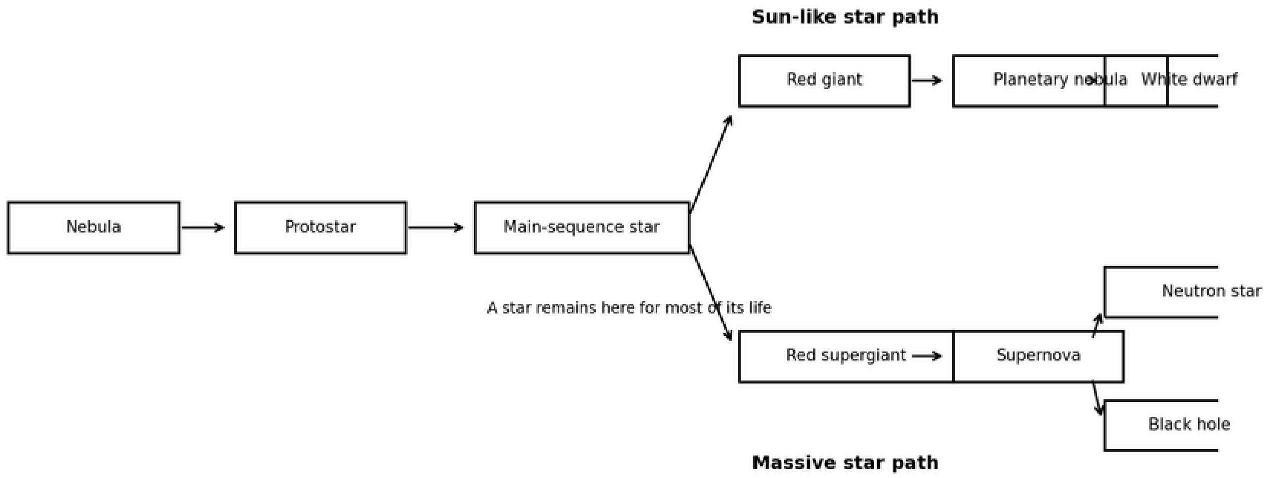

Figure 3. Simplified stellar-evolution pathways.

16. Study Figure 3. The first stage shown in star formation is a

A. black hole B. supernova

C. white dwarf D. nebula

17. Study Figure 3. A Sun-like star is expected to end as a

A. neutron star only B. black hole only

C. white dwarf D. red supergiant

18. Study Figure 3. A supernova is shown after the

A. red supergiant stage B. white dwarf stage

C. planetary nebula stage D. comet stage

19. Study Figure 3. Which two possible remnants can follow a supernova?

A. Neutron star and black hole B. White dwarf and planet

C. Nebula and asteroid D. Red giant and comet

20. A nebula is a cloud of

A. solid iron without gas B. gas and dust in space

C. dark matter visible to the eye D. only liquid water

21. An open cluster usually contains

A. only black holes

B. every galaxy in the Universe

C. one planet only

D. stars formed together in a loose group

22. The Milky Way contains

A. stars, gas, dust and dark matter B. only one star

C. no nebulae D. only the Solar System

23. The Big Bang model describes the Universe as having expanded from an early state that was

A. hot and dense B. made only of planets

C. inside Earth D. cold and empty

24. A black hole can be detected indirectly by observing

A. its bright solid surface

B. effects on nearby matter and light

C. ocean waves

D. sound travelling from inside it

25. Earth's mantle lies between the

A. troposphere and stratosphere B. Moon and Earth

C. inner and outer magnetosphere D. crust and core

26. The troposphere is the layer where

A. stars are formed B. the liquid outer core exists

C. satellites never pass D. most weather occurs

27. Auroras are linked to charged particles guided by Earth's

A. ocean currents B. magnetic field

C. plate boundaries only D. mountain ranges

28. Many lunar craters remain visible for a long time because the Moon has

A. heavy rainfall

B. thick forests

C. strong rivers

D. very little atmosphere and erosion

29. The leading explanation for the Moon's formation involves

A. a comet becoming hollow

B. a giant impact with early Earth

C. Earth splitting because of tides last century

D. a star exploding beside Earth yesterday

30. During a lunar eclipse, the Moon moves through

A. the Sun's atmosphere B. Mars's shadow

C. a comet tail D. Earth's shadow

31. A weather satellite can monitor

A. only distant galaxies B. only underground rocks

C. sound in empty space D. clouds and storms

32. Remote-sensing images taken at different wavelengths can help identify

A. the thoughts of people

B. sound frequency in vacuum

C. vegetation, water and land surfaces

D. the age of every rock exactly

33. A spacecraft thermal-control system helps keep equipment

A. within a safe temperature range B. outside every orbit

C. free from all motion D. invisible

34. A gyroscope or star tracker can help determine a spacecraft's

A. orientation B. fuel chemical formula only

C. launch date D. mass of Earth

35. The first stage of a multistage rocket is usually discarded after

A. its fuel is used B. the Sun sets

C. the payload lands D. the Moon becomes full

36. A heat shield is important for a returning spacecraft because re-entry produces

A. freezing only

B. zero air resistance at all altitudes

C. no frictional effects

D. intense heating

37. Mars Orbiter Mission was India's first interplanetary mission to

A. Venus B. Mercury

C. Mars D. Jupiter

38. Gaganyaan aims to demonstrate India's capability for

A. human spaceflight B. weather balloon racing

C. deep-sea mining only D. building optical fibres

39. A convex lens can form a real image on a screen when the object is

A. always touching the lens B. inside a black hole

C. placed beyond its focal point D. a radio wave

40. An electromagnet becomes stronger when

A. the wire is removed

B. current or number of coil turns is increased

C. the circuit is open

D. the iron core is replaced with air and current removed

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately difficult. Read carefully, calculate where needed and use elimination.

41. A star is 8 light-years away. The light received tonight left the star about

A. 8 days ago B. 8 years ago

C. 16 years ago D. 1 year ago

42. A telescope collects 400 units of light in 20 seconds. At the same rate, how much light will it collect in 75 seconds?

A. 3000 units B. 1200 units

C. 1500 units D. 750 units

43. Study Figure 3. Which statement is supported by the diagram?

A. Every star becomes a black hole

B. Initial mass affects the final stellar remnant

C. Nebulae form only after planets

D. White dwarfs form only after supernovae

44. A satellite camera records the same region every 5 days. Images are taken on 1 June, 6 June and 11 June. The next image should be on

A. 16 June B. 21 June

C. 15 June D. 14 June

45. A 10 N force and a 6 N force act in opposite directions on a cart. The net force is

A. 4 N towards the 6 N force B. 4 N towards the 10 N force

C. 16 N in both directions D. 60 N

46. A 3 kg satellite model moving at 2 m/s is stopped. The magnitude of its change in momentum is

A. 5 kg m/s B. 9 kg m/s

C. 6 kg m/s   D. 1.5 kg m/s

47. A student compares two insulating materials by wrapping identical warm bottles. Which measurement best identifies the better insulator?

A. The brighter colour only

B. The bottle with the larger label

C. The heavier wrapper regardless of cooling

D. The bottle that cools more slowly

48. A lunar eclipse is visible from a large part of Earth's night side, while a total solar eclipse is visible from a narrow path. The best reason is that

A. Earth has no atmosphere

B. the Moon stops orbiting during a lunar eclipse

C. the Sun becomes smaller at night

D. Earth's shadow at the Moon is much wider than the Moon's darkest shadow on Earth

49. A spacecraft sends a signal at 12:00:00 and receives a reply at 12:00:14. If processing time is negligible, the one-way travel time is

A. 7 s   B. 28 s

C. 14 s   D. 1 s

50. All black holes in a diagram are stellar remnants. Object X is not a stellar remnant. Which conclusion is logically valid?

A. X has no gravity   B. X must be a planet

C. X must emit visible light   D. X is not one of those black holes

MOCK PAPER 4

ISO 2026 Mock Paper - Intermediate Level (Classes VII-VIII)

Time: 90 minutes   Maximum Marks: 50

Name	Date	Score	Time taken
		/50	

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question. Suggested time: 90 minutes.

1. An observer's horizon is the apparent line where

A. the Moon meets Mars

B. the telescope meets a mirror

C. sky seems to meet the ground or sea

D. Earth meets the Sun

2. The celestial equator is the projection of Earth's

A. axis onto the ground B. equator onto the sky

C. orbit onto the Moon D. magnetic field into the ocean

3. A spectroscope separates light into

A. its component wavelengths B. sound frequencies only

C. gravitational waves only D. solid particles

4. Which instrument is best for collecting radio waves from space?

A. Thermometer B. Magnifying glass

C. Radio dish D. Barometer

5. Atmospheric turbulence makes stars appear to

A. stop producing light B. change mass rapidly

C. orbit Earth each minute D. twinkle

6. A telescope in space avoids

A. distortion and absorption by much of Earth's atmosphere

B. all need for power

C. all temperature control

D. all communication delay

7. The Sun contains most of the mass of the

A. Solar System B. Milky Way

C. entire Universe D. local galaxy cluster

8. The planets orbit in nearly the same plane mainly because they formed in a

A. flattened rotating disk B. straight line of stars

C. perfect cube D. stationary ocean

9. Saturn's rings are made mainly of

A. ice and rock particles B. hot gas only

C. continuous solid metal D. liquid oceans

10. Which planet has the strongest well-known global greenhouse heating?

A. Mars B. Mercury

C. Venus   D. Neptune

11. The coma of a comet is the

A. ring around Saturn

B. solid iron core of Earth

C. trail left by a rocket

D. cloud around its nucleus when heated

12. A small body that may pose an impact risk is carefully tracked as a

A. Near-Earth Object   B. white dwarf

C. geostationary satellite   D. main-sequence star

13. A star's colour gives information about its

A. distance with certainty   B. orbital speed around Earth

C. surface temperature   D. number of moons

14. A red giant has a larger radius than the star had during much of its

A. galaxy phase   B. main-sequence life

C. comet phase   D. planetary phase

15. A globular cluster usually contains many

A. old stars packed in a roughly spherical group

B. comet tails

C. new planets in a flat ring

D. communication satellites

16. A galaxy is held together mainly by

A. ocean tides   B. gravity

C. chemical glue   D. air pressure

17. Earth's inner core is mainly

A. solid iron and nickel   B. hydrogen gas

C. granite crust   D. liquid water

18. Greenhouse gases warm the lower atmosphere by

A. stopping Earth's rotation

B. absorbing and re-emitting infrared radiation

C. creating gravity

D. blocking all visible light

19. Earth observation from space is useful for monitoring

A. only the Moon's far side   B. only submarine sounds

C. only distant stars   D. forests, crops, floods and cities

20. Space weather can disturb satellites because it involves

A. ocean tides only

B. Earthquakes on Mars only

C. rainfall inside spacecraft

D. solar activity and charged particles

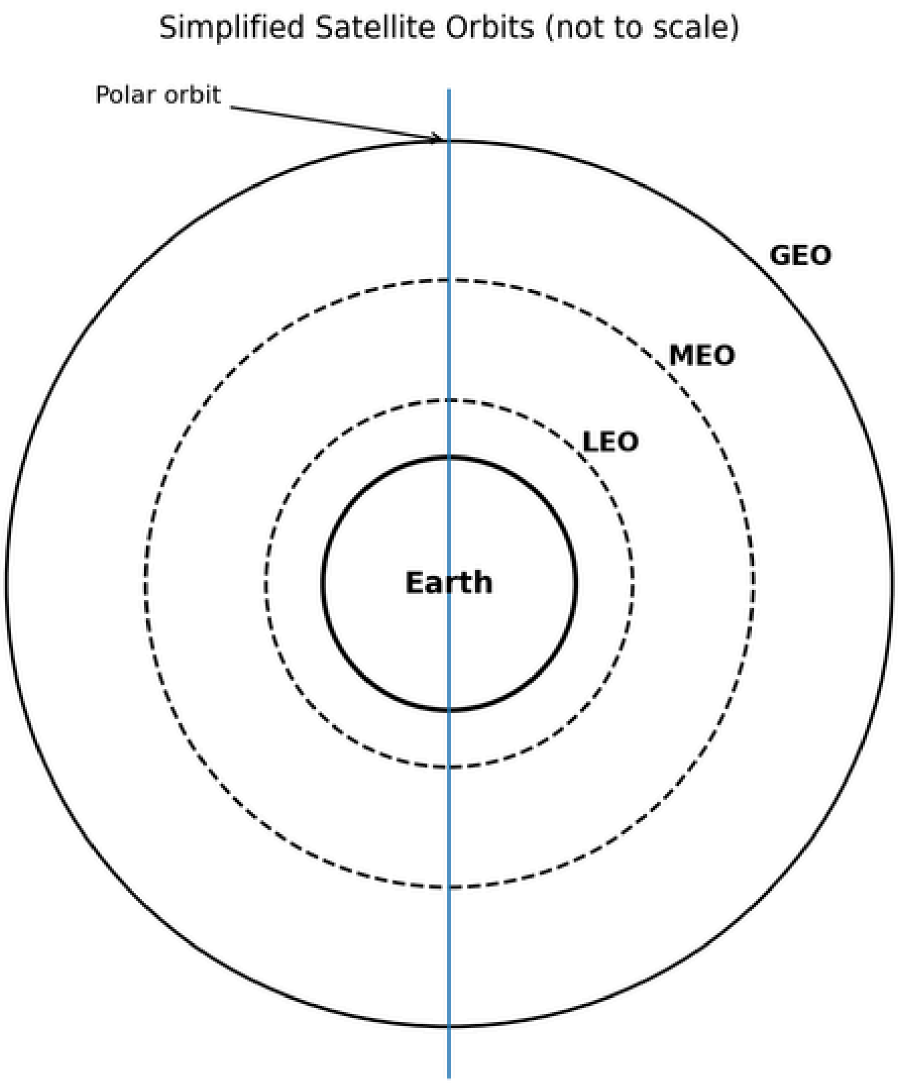

Figure 4. Simplified satellite orbits; not to scale.

21. Study Figure 4. Which orbit is closest to Earth?

A. LEO B. GEO

C. Polar orbit always farthest D. MEO

22. Study Figure 4. Navigation constellations commonly use

A. a comet orbit B. the Earth's surface

C. MEO D. the Moon's core

23. Study Figure 4. A satellite that remains above nearly the same longitude uses

A. a random suborbital path B. GEO

C. an escape path D. a low polar orbit

24. Study Figure 4. A polar orbit passes near

A. both poles B. Jupiter and Saturn

C. the Sun only D. the equator only

25. A satellite bus is the main structure that supports

A. payload and spacecraft subsystems B. a railway engine

C. a ground telescope dome only D. an ocean ship

26. The payload of an Earth-observation satellite may include a

A. launch tower B. parachute for every orbit

C. fuel refinery D. camera or imaging sensor

27. Telemetry is data sent from a spacecraft to

A. only another star

B. the ocean floor without a receiver

C. the centre of Earth

D. ground controllers

28. A rocket engine produces thrust by ejecting mass at high speed

A. without any direction B. backward

C. sideways only D. forward only

29. A cryogenic rocket engine uses propellants stored at

A. room temperature only B. very low temperatures

C. no temperature D. the Sun's temperature

30. Orbital insertion is the manoeuvre used to

A. land a telescope in the ocean

B. turn a planet into a moon

C. place a spacecraft into its intended orbit

D. remove all gravity

31. A launch pad includes systems for

A. support, fuelling, electrical connections and safety

B. forming clouds

C. changing Earth's mass

D. making stars

32. PSLV, GSLV and LVM3 are all Indian

A. astronaut suits B. space stations

C. launch vehicles D. Moon rovers

33. GSLV uses a cryogenic stage to help place payloads into

A. ocean currents B. higher-energy orbits

C. the Sun's core D. underground tunnels

34. Chandrayaan missions explore the

A. Sun B. Mars

C. Venus D. Moon

35. Aditya-L1 operates near the Sun-Earth

A. North Pole B. L1 region

C. Moon's surface D. asteroid belt centre

36. The main purpose of SpaDeX is important for future missions requiring

A. faster sound in vacuum B. docking and assembly in orbit

C. underwater navigation D. weather balloons

37. The ISS appears to have microgravity because the station and its contents are

A. outside Earth's gravity completely

B. motionless

C. falling together around Earth

D. supported by air

38. A space telescope can observe wavelengths blocked by Earth's atmosphere, such as much of the

A. X-ray radiation B. visible green light only

C. ocean-wave spectrum   D. sound spectrum

39. A 50 N force acts for 4 seconds. The impulse is

A. 12.5 N s   B. 46 N s

C. 200 N s   D. 54 N s

40. A concave mirror can focus parallel light rays at its

A. horizon   B. equator

C. focus   D. magnetic pole

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately difficult. Read carefully, calculate where needed and use elimination.

41. A 1000 kg spacecraft ejects gas backward. Immediately after ejection, the spacecraft gains forward momentum. This demonstrates conservation of

A. temperature   B. volume

C. brightness   D. momentum

42. Study Figure 4. A satellite in LEO completes many orbits while Earth rotates beneath it. This helps it

A. avoid passing over oceans

B. stop moving relative to space

C. remain over one city forever

D. cover different ground tracks over time

43. A rocket has a total mass of 5000 kg and produces a net upward force of 10000 N. Its upward acceleration is

A. 5 m/s^2 B. 0.5 m/s^2

C. 50 m/s^2 D. 2 m/s^2

44. A satellite moves in a circular orbit at constant speed. Which statement is correct?

A. Its momentum is always zero

B. Its velocity changes because direction changes

C. Its acceleration is zero

D. No force acts on it

45. A 2 kg object is lifted vertically by 5 m. Taking $g = 10\ m/s^2$, its gain in gravitational potential energy is

A. 10 J   B. 20 J

C. 100 J   D. 50 J

46. A satellite image has 10 m spatial resolution. Compared with a 100 m-resolution image, it can generally show

A. less detail only   B. smaller ground details

C. no land features D. only weather

47. Mission A needs continuous communication with one region. Mission B needs global mapping. The best pair is

A. A: escape orbit; B: landing path

B. A: lunar orbit; B: no orbit

C. A: polar; B: geostationary

D. A: geostationary; B: polar

48. A student concludes that black paper heats faster than white paper after one trial. What is the best improvement?

A. Ignore measurement uncertainty

B. Change every variable at once

C. Use different thermometers without checking them

D. Repeat trials and control material, size and starting temperature

49. A spacecraft command takes 12 minutes to reach a planet. If the spacecraft immediately replies, the earliest reply reaches Earth after

A. 6 min B. 12 min

C. 24 min D. 18 min

50. Three statements are given: (1) All GEO satellites orbit above the equator. (2) Satellite R is in GEO. (3) Satellite S passes over both poles. Which conclusion is certain?

A. S is also in GEO

B. R orbits above the equator

C. R and S are the same satellite

D. S remains over one longitude

MOCK PAPER 5

ISO 2026 Mock Paper - Intermediate Level (Classes VII-VIII)

Time: 90 minutes Maximum Marks: 50

Name	Date	Score	Time taken
		/50	

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question. Suggested time: 90 minutes.

1. The daily east-to-west motion of stars across the sky is mainly caused by Earth's

A. tides

B. revolution around the Sun each day

C. rotation from west to east

D. magnetic field

2. The north celestial pole lies near the direction of

A. Jupiter at all times

B. the Moon at full phase

C. the Sun at noon every day

D. Polaris for observers in the Northern Hemisphere

3. An astronomer measures the angle between two stars. This is an

A. pressure measurement B. angular measurement

C. mass measurement D. temperature measurement

4. A CCD in a telescope is used mainly to

A. produce rocket thrust

B. record incoming light electronically

C. create radio waves in stars

D. measure ocean depth by touch

5. Which sequence correctly orders electromagnetic waves from lower to higher frequency?

A. Radio, infrared, visible, ultraviolet

B. Gamma, X-ray, visible, radio

C. Visible, radio, gamma, infrared

D. Ultraviolet, visible, infrared, radio

6. A redshift in a galaxy's spectrum usually indicates that the galaxy is

A. moving towards us B. inside the Solar System

C. not moving at all D. moving away from us

7. The rocky planets formed mainly in the warmer inner Solar System because

A. gravity did not exist farther out

B. the Sun formed after the planets

C. all gas was frozen there

D. rock and metal could condense there more easily than volatile ices

8. Planetary motion is governed mainly by the gravity of the

A. Milky Way centre alone B. Sun

C. Moon only D. asteroid belt

9. Jupiter and Saturn are classified as

A. gas giants B. dwarf planets

C. comets D. terrestrial planets

10. A planet with a thick carbon-dioxide atmosphere can experience strong

A. greenhouse warming

B. loss of all heat

C. ocean freezing caused by no atmosphere

D. magnetic cooling only

11. A comet nucleus contains mixtures of

A. only hydrogen gas B. liquid oceans and forests

C. only molten iron D. ice, dust and rock

12. Objects in the Kuiper Belt orbit the Sun mostly

A. beyond Neptune B. inside Mercury's orbit

C. between Earth and Mars D. inside Earth

13. A protostar forms when part of a nebula

A. collapses under gravity B. becomes a planet instantly

C. stops containing gas D. moves outside the Universe

14. The main factor deciding whether a star ends as a white dwarf or after a supernova is its

A. initial mass B. distance from Earth

C. apparent brightness only D. number of planets

15. The centre of many large galaxies may contain a

A. solid wall B. single asteroid only

C. supermassive black hole D. large ocean

16. Cosmic expansion means that, on very large scales, distances between many galaxies are

A. equal to Earth-Moon distance B. decreasing to zero

C. increasing D. unchanged everywhere

17. The lithosphere includes Earth's crust and the rigid upper part of the

A. inner core   B. mantle

C. stratosphere   D. outer core

18. Earth's magnetic field is generated mainly by motion in the

A. oceans only   B. solid crust only

C. liquid outer core   D. troposphere

19. A full Moon occurs when

A. the Moon produces extra light

B. the Sun is between Earth and Moon

C. Earth is approximately between the Sun and Moon

D. the Moon is between Sun and Earth

20. Neap tides occur near the

A. solar eclipse only   B. new and full Moon only

C. lunar eclipse only   D. first and third quarter phases

21. A satellite in low Earth orbit experiences slight atmospheric drag because

A. there is no atmosphere at any altitude

B. gravity stops in LEO

C. the upper atmosphere is extremely thin but not completely absent

D. the satellite is motionless

22. A navigation receiver finds its position by comparing

A. signals and travel times from several satellites

B. sound waves from the Moon

C. only one weather photograph

D. the colour of stars

23. A satellite antenna must often be pointed accurately because

A. radio communication uses directed links

B. all signals are visible light

C. the antenna changes the orbit by itself

D. gravity travels through the antenna

24. Onboard computers help a spacecraft to

A. remove all heat without radiators

B. process commands and control systems

C. stop orbital motion

D. create infinite power

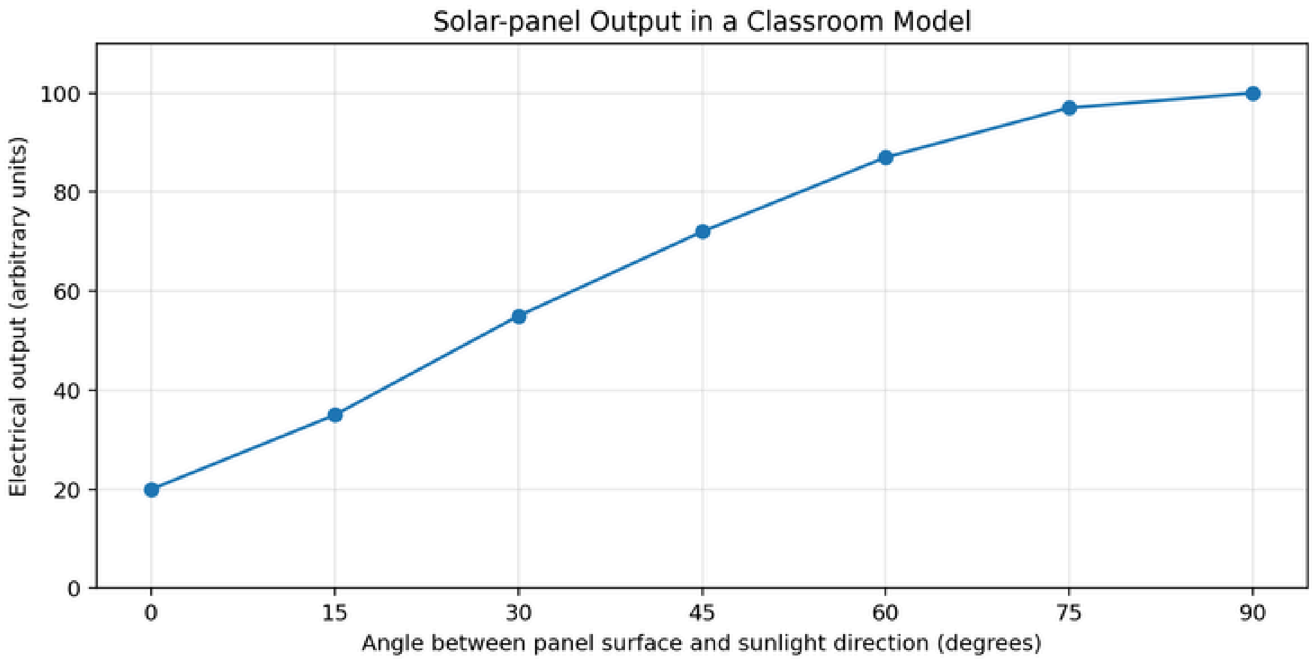

Figure 5. Output of a classroom solar-panel model.

25. Study Figure 5. The electrical output increases most rapidly between

A. 0° and 15° B. 30° and 45°

C. 75° and 90° D. 15° and 30°

26. Study Figure 5. At which tested angle is the output highest?

A. 30° B. 90°

C. 60° D. 0°

27. Study Figure 5. The graph suggests the model panel works best when its surface is

A. cooled with ice only B. hidden behind a wall

C. more nearly facing the light D. parallel to the light rays

28. The high speed of rocket exhaust is produced in a

A. parachute canopy B. rocket engine nozzle

C. radio dish D. solar panel

29. A rocket stage that has finished burning is separated mainly to

A. stop the payload entering space

B. increase drag

C. reduce the mass that later stages must accelerate

D. make the rocket shorter for decoration

30. A reusable booster must perform controlled guidance and landing so that it can

A. remain as space debris

B. avoid using engines

C. carry no fuel

D. be recovered and flown again after inspection

31. Which launch vehicle launched Aditya-L1?

A. LVM3 B. PSLV

C. GSLV Mk II D. Sounding rocket

32. Which mission studies polarisation of X-rays from astronomical sources?

A. Gaganyaan B. Mangalyaan

C. Chandrayaan-1 D. XPoSat

33. Which mission demonstrated docking between two Indian spacecraft in 2025?

A. INSAT-3D B. RISAT

C. SpaDeX D. AstroSat

34. Which mission is planned to demonstrate Indian human spaceflight capability?

A. Gaganyaan B. XPoSat

C. Chandrayaan-1 D. Aditya-L1

35. The James Webb Space Telescope mainly observes the Universe in

A. sound waves B. infrared wavelengths

C. ocean waves D. only radio broadcasts

36. A Mars rover uses cameras and instruments because scientists cannot

A. send radio signals to Mars

B. measure rocks remotely

C. control it by direct human presence on Mars

D. use electrical power

37. Newton's first law is also called the law of

A. inertia B. pressure

C. heat conduction   D. reflection

38. For the same force, an object with smaller mass has

A. the same acceleration as every object

B. greater acceleration

C. zero acceleration

D. smaller acceleration always

39. A ray of light changes direction when it passes from air into glass. This is

A. refraction   B. magnetism

C. evaporation   D. reflection only

40. An electric current in a wire produces a

A. gravitational shield   B. magnetic field

C. vacuum   D. comet tail

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately difficult. Read carefully, calculate where needed and use elimination.

41. Study Figure 5. Estimate the output at 45° as a percentage of the output at 90°.

A. About 145%   B. About 50%

C. About 27%   D. About 72%

42. A satellite makes 15 orbits in 24 hours. Its average orbital period is

A. 360 h   B. 9 h

C. 0.625 h   D. 1.6 h

43. A rocket engine gives a constant thrust of 6000 N to a 2000 kg vehicle after gravity and drag are already accounted for. The acceleration is

A. 0.33 m/s^2 B. 12,000 m/s^2

C. 8 m/s^2 D. 3 m/s^2

44. A 5 kg model rocket moving at 12 m/s has kinetic energy of

A. 360 J   B. 60 J

C. 720 J   D. 30 J

45. An optical image of a star spreads over 4 pixels, while a second image spreads over 12 pixels under the same conditions. Which image is more sharply focused?

A. Sharpness cannot relate to image spread

B. Both are always identical

C. The 4-pixel image

D. The 12-pixel image

46. A researcher wants to test whether telescope exposure time changes measured brightness. Which plan is fairest?

A. Change exposure and aperture together

B. Compare observations from different weather without recording it

C. Use different stars and different telescopes each time

D. Use the same star, telescope and settings while changing only exposure time

47. A spacecraft receives 900 W from solar panels. Instruments use 420 W, heaters use 180 W and communications use 150 W. How much power remains?

A. 100 W B. 150 W

C. 250 W D. 50 W

48. If the Sun suddenly disappeared in a thought experiment, Earth would continue along the tangent to its orbit because of

A. pressure B. inertia

C. magnetism only D. tides

49. A graph of distance against time is a straight line with constant positive slope. This indicates

A. changing direction every second B. increasing acceleration only

C. constant speed D. no motion

50. Statements: Every mission in Set M uses remote sensing. Mission T does not use remote sensing. Which must be true?

A. T is not in Set M B. Set M is empty

C. T is a communication satellite D. T is in Set M

Answer Key

Use this section after completing a paper. The Olympiad Challenge explanations follow the compact answer tables.

Mock Paper 1

1	2	3	4	5	6	7	8	9	10
D	B	A	B	C	A	D	C	C	D
11	12	13	14	15	16	17	18	19	20
C	B	B	C	C	D	B	C	A	A
21	22	23	24	25	26	27	28	29	30
D	C	A	A	A	B	A	B	B	B
31	32	33	34	35	36	37	38	39	40
C	B	D	C	D	D	A	D	B	A
41	42	43	44	45	46	47	48	49	50
C	A	D	D	A	B	C	A	B	D

Mock Paper 2

1	2	3	4	5	6	7	8	9	10
D	B	B	D	A	D	C	B	C	D
11	12	13	14	15	16	17	18	19	20
A	B	C	A	B	A	A	D	A	C
21	22	23	24	25	26	27	28	29	30
D	B	D	A	D	B	D	C	B	A
31	32	33	34	35	36	37	38	39	40
D	B	B	C	B	A	A	D	C	C
41	42	43	44	45	46	47	48	49	50
C	D	A	A	C	A	C	B	C	B

Mock Paper 3

1	2	3	4	5	6	7	8	9	10
B	C	C	C	C	A	B	B	A	A
11	12	13	14	15	16	17	18	19	20
D	B	C	C	B	D	C	A	A	B
21	22	23	24	25	26	27	28	29	30
D	A	A	B	D	D	B	D	B	D
31	32	33	34	35	36	37	38	39	40
D	C	A	A	A	D	C	A	C	B
41	42	43	44	45	46	47	48	49	50
B	C	B	A	B	C	D	D	A	D

Mock Paper 4

1	2	3	4	5	6	7	8	9	10
C	B	A	C	D	A	A	A	A	C
11	12	13	14	15	16	17	18	19	20
D	A	C	B	A	B	A	B	D	D
21	22	23	24	25	26	27	28	29	30
A	C	B	A	A	D	D	B	B	C
31	32	33	34	35	36	37	38	39	40
A	C	B	D	B	B	C	A	C	C
41	42	43	44	45	46	47	48	49	50
D	D	D	B	C	B	D	D	C	B

Mock Paper 5

1	2	3	4	5	6	7	8	9	10
C	D	B	B	A	D	D	B	A	A
11	12	13	14	15	16	17	18	19	20
D	A	A	A	C	C	B	C	C	D
21	22	23	24	25	26	27	28	29	30
C	A	A	B	D	B	C	B	C	D
31	32	33	34	35	36	37	38	39	40
B	D	C	A	B	C	A	B	A	B
41	42	43	44	45	46	47	48	49	50
D	D	D	A	C	D	B	B	C	A

Olympiad Challenge Explanations

Short explanations are provided for Questions 41-50 of every paper. Students should redo each calculation without looking at the answer.

Mock Paper 1: Questions 41-50

41. C - The signal needs 1.3 s each way, so total time is 2.6 s.

42. A - Altitude is measured upward from the horizon, while azimuth/direction identifies where along the horizon the object lies.

43. D - Outer planets travel along larger orbits and generally have longer orbital periods.

44. D - Pressure = force/area = 120/0.03 = 4000 Pa.

45. A - The reading remains 35 units from 09:00 to 10:00.

46. B - The tilted lunar orbit means the three bodies are usually not exactly aligned.

47. C - Mass does not depend on location, but weight depends on local gravity.

48. A - A larger aperture gathers more light and improves angular resolution.

49. B - A fair test changes colour while controlling other important factors.

50. D - The repeating group is star, planet, planet.

Mock Paper 2: Questions 41-50

41. C - Average speed = 240/8 = 30 m/min.

42. D - Earth is 1 AU and Jupiter is 5.2 AU.

43. A - For circular orbits, a larger radius gives a larger circumference.

44. A - Magnification = objective focal length/eyepiece focal length = 1000/20 = 50.

45. C - Both are at room temperature, but metal transfers heat faster.

46. A - Momentum = mass × velocity = 2 × 4 = 8 kg m/s.

47. C - The measured outcome is current.

48. B - GPS constellations include multiple satellites, allowing redundancy and geometry for positioning.

49. C - Massive stars form in nebulae, spend most of life on the main sequence, expand, then may explode.

50. B - This follows directly from the rule and membership of P in the list.

Mock Paper 3: Questions 41-50

41. B - A light-year is the distance light travels in one year, so we see the star as it was 8 years earlier.

42. C - Rate = 400/20 = 20 units/s; 20 × 75 = 1500.

43. B - The diagram separates Sun-like and massive-star paths.

44. A - The sequence increases by 5 days each time.

45. B - Opposite forces subtract: 10 - 6 = 4 N.

46. C - Initial momentum = 3 × 2 = 6; final is zero.

47. D - A better insulator reduces heat transfer, so temperature falls more slowly.

48. D - The geometry produces a broad terrestrial shadow at lunar distance and a narrow umbral track on Earth.

49. A - Round trip is 14 s, so one way is 7 s.

50. D - If every item in the stated black-hole set is a remnant, a non-remnant cannot belong to that set.

Mock Paper 4: Questions 41-50

41. D - The total momentum of the spacecraft-plus-exhaust system is conserved.

42. D - Earth rotates under the orbit, so successive passes cover different regions.

43. D - a = F/m = 10000/5000 = 2 m/s^2.

44. B - Velocity includes direction; centripetal acceleration changes the direction continuously.

45. C - mgh = 2 × 10 × 5 = 100 J.

46. B - Smaller resolution distance means finer spatial detail.

47. D - GEO supports continuous regional coverage; polar passes support global mapping.

48. D - Repeated controlled trials improve reliability.

49. C - The round trip takes 12 + 12 = 24 minutes.

50. B - Statement 1 applied to R gives the conclusion.

Mock Paper 5: Questions 41-50

41. D - The graph gives about 72 units at 45° and 100 at 90°, so about 72%.

42. D - Period = 24/15 = 1.6 hours.

43. D - a = F/m = 6000/2000 = 3 m/s^2.

44. A - KE = 1/2 mv^2 = 0.5 × 5 × 144 = 360 J.

45. C - A more compact point image indicates better focus under the same conditions.

46. D - Only the independent variable should change in a controlled test.

47. B - 900 - 420 - 180 - 150 = 150 W.

48. B - Without centripetal gravity, inertia would carry Earth approximately along a straight tangent.

49. C - A constant slope on a distance-time graph represents constant speed.

50. A - By contradiction, T cannot be a member of a set whose every member uses remote sensing.

Practice Score Guide

Score	Interpretation	Recommended action
45-50	Excellent command	Review only errors and attempt more timed papers.
38-44	Strong preparation	Revise the weakest two syllabus units.
30-37	Developing well	Relearn concepts behind incorrect answers before retesting.
20-29	Needs structured revision	Study one unit at a time and practise basic calculations.
Below 20	Foundation needs strengthening	Use the Intermediate textbook with teacher or parent guidance before another full mock.